

Core Concepts

What is the difference between generative models and discriminative models? Why do we need generative models? Discriminative models learn the decision boundary to classify data, while generative models learn the data distribution to generate new samples 1. We need generative models specifically to create or synthesize new data 1.

What is the difference between a foundational model and a traditional machine learning model? The sources do not explicitly define the difference between them, though they note foundation models involve large-scale pretraining (like language modeling) followed by fine-tuning, whereas traditional models often train on specific tasks like ImageNet 2.

What are diffusion models and how do they function? Diffusion models use a forward process that gradually adds noise to an image via a Markov chain until it is pure Gaussian noise 3. They then use a reverse process where a U-Net predicts and subtracts that noise to recover a clean image 4.

What is the difference between VAEs (Variational Autoencoders) and Stable Diffusion? VAEs encode inputs into a smooth probability distribution to generate samples via a reparameterization trick 5. Stable Diffusion is a Latent Diffusion Model (LDM) that runs a noise-adding/subtracting diffusion process within a compressed VAE latent space and uses cross-attention for text conditioning 4.

How do autoencoders work? Autoencoders work by passing an input through an encoder to compress it into a latent space (z), and then using a decoder to reconstruct the original input from that latent space 6.

GAN & VAE

What is the difference between GANs and Variational Autoencoders (VAEs)? GANs use a minimax game between a generator creating fake samples and a discriminator trying to detect them 7. VAEs encode inputs into a continuous distribution and are trained by minimizing reconstruction loss and KL divergence 5, 8.

What is mode collapse in GANs? The sources list mode collapse as one of the three common challenges in GAN training (alongside instability and vanishing gradients), but they do not explicitly define what it is 7.

Attention Mechanisms

Why do we need the attention mechanism? Attention allows models like Transformers to process data in parallel and better handle long-range dependencies, overcoming the limitations of sequential RNNs 8.

Explain the attention mechanism and how it works. Attention computes weighted scores using Queries (what I'm looking for), Keys (what I contain), and Values (what I return if matched) 8. It uses the formula $\text{Softmax}(QK^T/\sqrt{d_k}) \cdot V$ to stabilize variance and output relevant context 9, 10.

What is the difference between self-attention and cross-attention?In **self-attention**, the **Query, Key, and Value** all come from the same sequence 11. In **cross-attention**, the **Query** comes from one source, while the **Key** and **Value** come from a different source 11.

What is the difference between multi-head attention and cross-attention?Multi-head attention involves **running multiple attention operations in parallel** with different learned projections 10. Cross-attention specifically refers to the routing of inputs, where the **Query** comes from a different sequence than the **Keys** and **Values** 11.

What is multi-head attention (MHA)?Multi-head attention involves **running multiple attention operations in parallel** using different learned projections, and then concatenating their outputs 10.

What is multi-query attention (MQA)?The provided sources and conversation history do not contain any information about multi-query attention (MQA).

What is masked attention?Masked attention, or causal masking, **prevents the model from attending to future tokens** during autoregressive text generation 10.

What is the difference between attention and Flash Attention?Standard attention is highly memory-bound with $O(N^2)$ memory usage 12. **Flash Attention uses tiling and recomputation** in SRAM to drastically reduce memory traffic, resulting in a **3× speedup while returning exact outputs** 13.

Transformers & Embeddings

Why do we need positional encoding?Positional encoding is needed to **preserve spatial or sequential information** when combining features, since Transformers process inputs in parallel rather than sequentially 3, 8.

What is positional encoding?Positional encoding (such as sinusoidal functions) is a technique added to inputs in Transformers to **preserve spatial information** and order, though the sources do not detail its exact mathematical mechanics 3, 9.

What is the difference between one-hot encoding and embeddings?One-hot encoding creates **large, sparse matrices with high memory costs and no semantic meaning** 14.

Embeddings are dense vector representations that capture semantic context and drastically reduce memory needs 14, 15.

RNN

What are the shortcomings of RNNs?RNNs are limited by their **sequential nature, lack of parallelization, and difficulty in handling long-range dependencies** 8, 16.

Do RNNs process tokens sequentially or in parallel?RNNs process tokens **sequentially** 8, 16.

Fine-Tuning & Optimization

What is LoRA (Low-Rank Adaptation)?LoRA **approximates weight updates using low-rank matrices** ($\Delta W = BA$), which drastically reduces the number of trainable parameters while keeping the base model frozen 17, 18.

What is quantization of LoRA?Quantization involves **converting weights from FP32 to Int8 or 4-bit precision** using block-wise scaling to handle outliers, significantly reducing memory usage 19, 20.

What is the difference between LoRA and QLoRA?LoRA updates weights using low-rank matrices on a standard base model 18. **QLoRA combines a 4-bit quantized base model** with BF16 LoRA adapters and on-the-fly dequantization to fine-tune massive models on minimal memory 20.

What are different fine-tuning methods for models?Methods include **Supervised Fine-Tuning (SFT/instruction tuning)**, **Preference Tuning (RLHF/DPO)**, full fine-tuning, and Parameter-Efficient methods like **Adapters, LoRA, and QLoRA** 2, 19.

What is efficient fine-tuning in the context of LLMs?Efficient fine-tuning uses techniques like **Adapters or LoRA to update only a tiny fraction of parameters** (e.g., 0.01%), avoiding the slow training and high storage costs of full fine-tuning 2, 18, 21.

What is PEFT (Parameter-Efficient Fine-Tuning)?PEFT refers to methods that mitigate the high costs of full fine-tuning by **updating only a small number of parameters**, using bottleneck adapters or low-rank matrices 2, 21.

Hallucination & RAG

What are hallucinations in LLMs?The sources note that techniques like RAG help reduce hallucinations, but they **do not explicitly define** what hallucinations are 22.

What are techniques to avoid hallucinations?The primary technique mentioned to avoid hallucinations is **Retrieval-Augmented Generation (RAG)** 22, 23.

How can hallucinations be mitigated in LLMs?Hallucinations can be mitigated by **using the RAG pipeline**, which grounds the model's responses in factual data retrieved from a vector database 22, 24.

What is RAG (Retrieval-Augmented Generation)?RAG is a pipeline that **augments prompts with factual data** by performing dense retrieval from a vector database, re-ranking the results, and generating an answer with citations 24.

How does RAG help in reducing hallucinations?RAG reduces hallucinations by **using dense retrieval to augment prompts with factual, grounded data** before the model generates its final answer 22, 23.